

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change]	84532	78.296469
1	[Git, Commit]	10995	10.183950
2	[File, Git]	5641	5.224890
3	[Git, Tag]	1591	1.473639
4	[Author, Git, Person]	1249	1.156867
5	[Json, Key]	668	0.618725
6	[Json, Value, Scalar]	603	0.558520
7	[Committer, Git, Person]	370	0.342707
8	[NPM, Dependency]	338	0.313067
9	[Type, TS, Primitive]	291	0.269534
10	[Type, TS, Declared]	277	0.256567
11	[TS, ExternalDeclaration]	215	0.199140
12	[Type, TS, Literal]	136	0.125968
13	[Json, Value, Object]	133	0.123189
14	[Type, TS, Union]	119	0.110222
15	[Type, TS, ObjectMember]	102	0.094476
16	[NPM, Script]	91	0.084287
17	[TS, Property]	65	0.060205
18	[TS, Function]	47	0.043533
19	[Type, TS, FunctionParameter]	40	0.037049
20	[Type, Object, TS]	39	0.036123
21	[Git, Branch]	38	0.035197
22	[File, Directory]	34	0.031492
23	[Type, TS, Function]	34	0.031492
24	[TS, Parameter]	33	0.030566
25	[Package, File, Json, NPM]	29	0.026861
26	[TS, Variable]	24	0.022230
27	[Value, TS, Literal]	20	0.018525
28	[jqAssistant, Rule, Concept]	19	0.017598
29	[Type, TS, Intersection]	17	0.015746

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

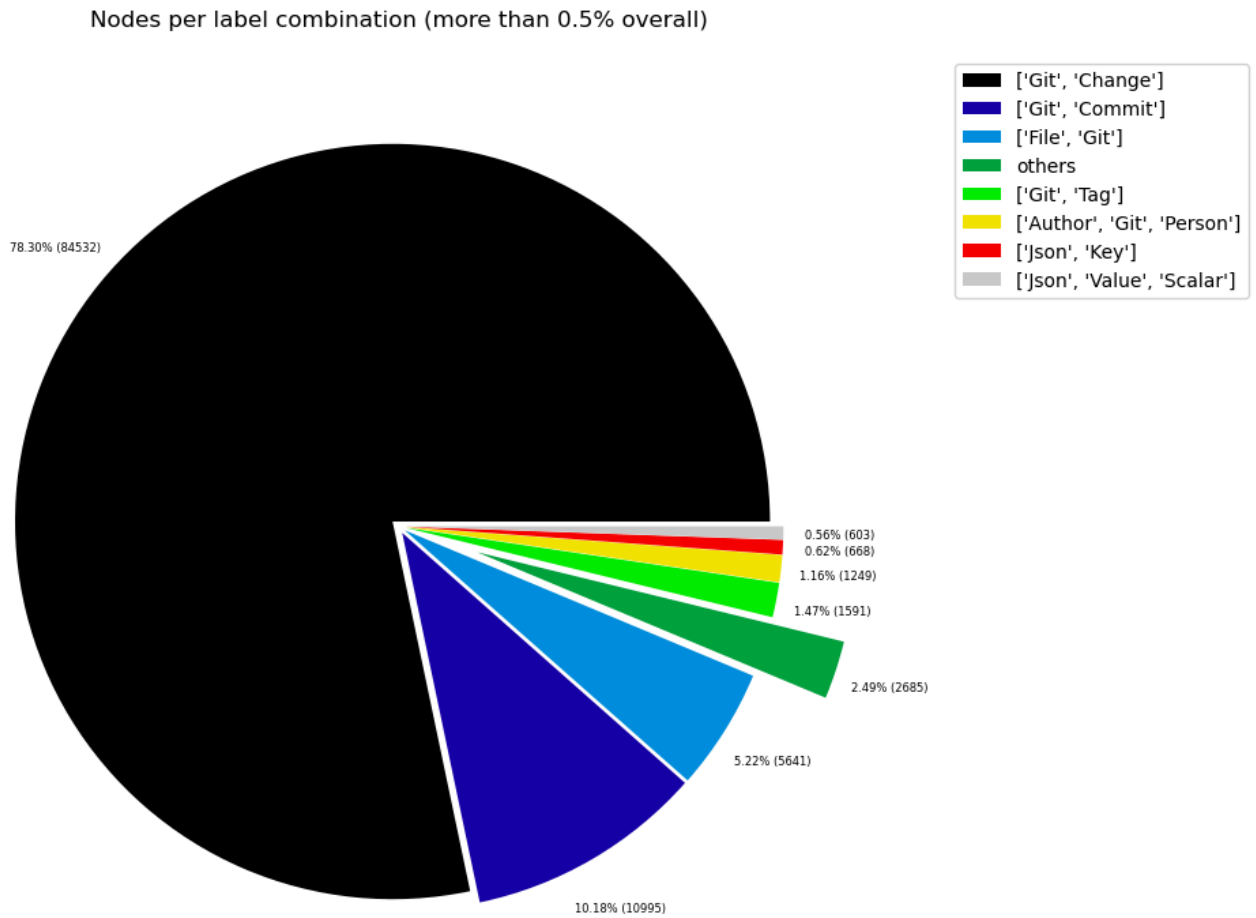


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000926
1	[File, TS, Scan]	1	0.000926
2	[TS, Method]	1	0.000926
3	[Repository, File, Git]	1	0.000926
4	[TS, Constructor]	1	0.000926
5	[Value, TS, ObjectMember]	1	0.000926
6	[TS, Class]	1	0.000926
7	[TS, Enum]	2	0.001852
8	[Value, Object, TS]	3	0.002779
9	[Type, TS, Tuple]	3	0.002779
10	[Value, TS, Function]	4	0.003705
11	[TS, TypeParameter]	4	0.003705
12	[Value, TS, Complex]	5	0.004631
13	[NPM, Engine]	6	0.005557
14	[Project, TS]	6	0.005557
15	[File, Local]	6	0.005557
16	[Value, TS, Call]	6	0.005557
17	[Value, TS, Member]	6	0.005557
18	[File, TS, Local, Module]	6	0.005557
19	[Type, TS, TypeParameterReference]	6	0.005557
20	[TS, EnumMember]	8	0.007410
21	[Type, TS, NotIdentified]	11	0.010189
22	[TS, ExternalModule]	11	0.010189
23	[Json, Value, Array]	12	0.011115
24	[Value, TS, Declared]	13	0.012041
25	[TS, TypeAlias]	16	0.014820
26	[File, Directory, Local]	16	0.014820
27	[Type, TS, Intersection]	17	0.015746
28	[TS, Interface]	17	0.015746
29	[JQAssistant, Rule, Concept]	19	0.017598

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

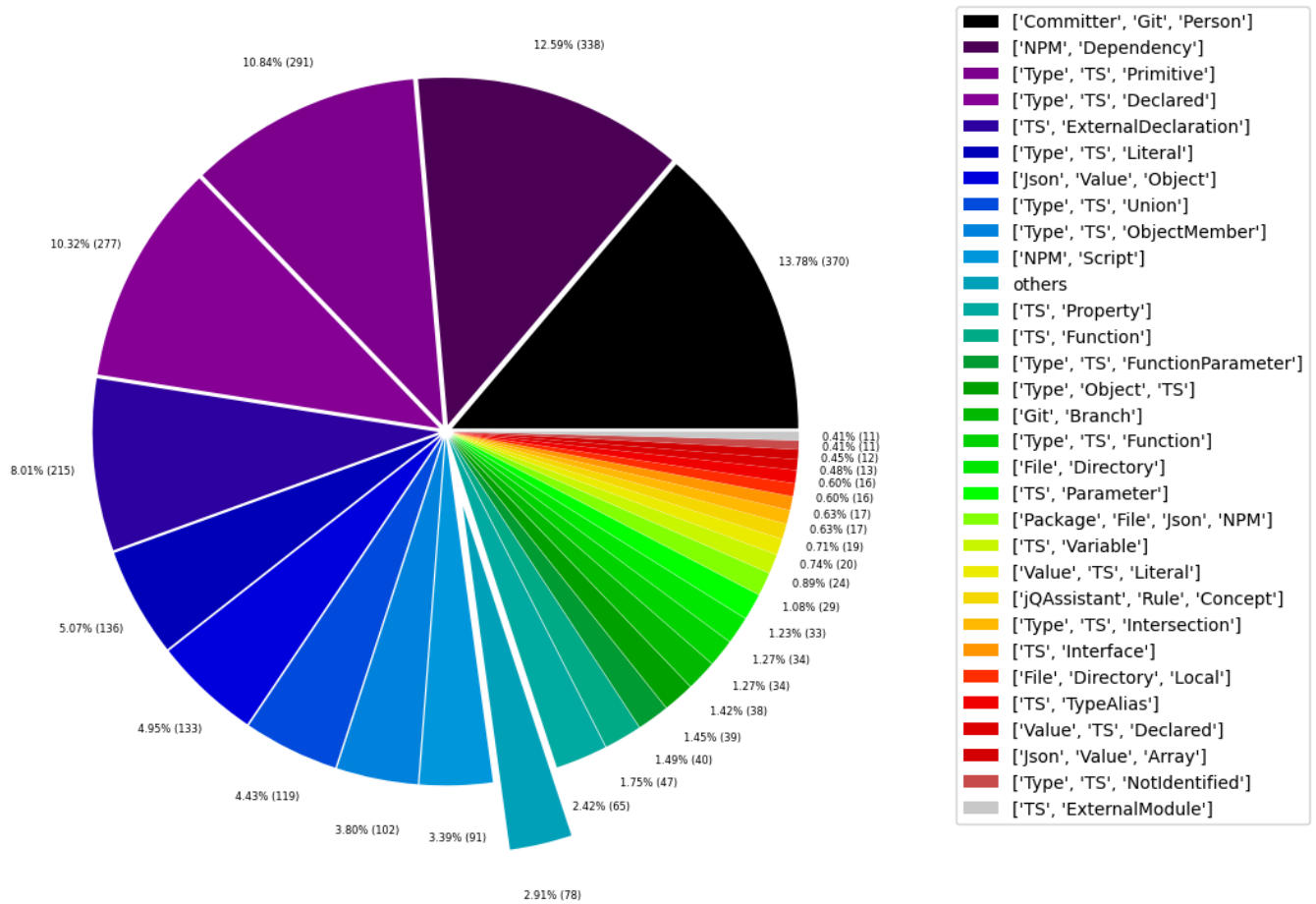


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

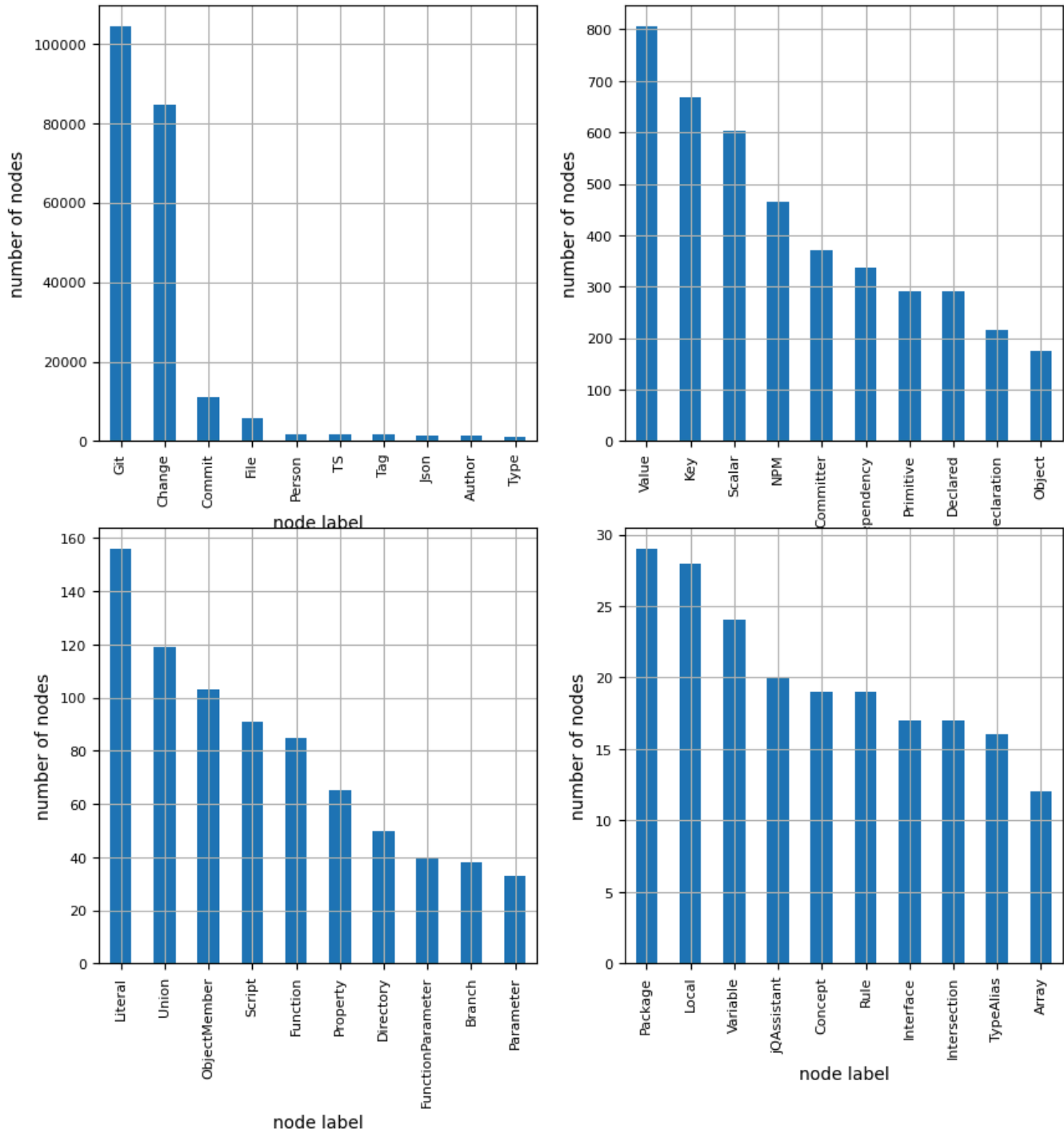
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	104417	96.714646
1	Change	84532	78.296469
2	Commit	10995	10.183950
3	File	5734	5.311030
4	Person	1619	1.499574
5	TS	1591	1.473639
6	Tag	1591	1.473639
7	Json	1445	1.338409
8	Author	1249	1.156867
9	Type	1075	0.995702
10	Value	806	0.746545
11	Key	668	0.618725
12	Scalar	603	0.558520
13	NPM	464	0.429773
14	Committer	370	0.342707
15	Dependency	338	0.313067
16	Primitive	291	0.269534
17	Declared	290	0.268608
18	ExternalDeclaration	215	0.199140
19	Object	175	0.162091
20	Literal	156	0.144493
21	Union	119	0.110222
22	ObjectMember	103	0.095402
23	Script	91	0.084287
24	Function	85	0.078730
25	Property	65	0.060205
26	Directory	50	0.046312
27	FunctionParameter	40	0.037049
28	Branch	38	0.035197
29	Parameter	33	0.030566
30	Package	29	0.026861
31	Local	28	0.025935
32	Variable	24	0.022230
33	jqAssistant	20	0.018525
34	Concept	19	0.017598
35	Rule	19	0.017598
36	Interface	17	0.015746
37	Intersection	17	0.015746
38	TypeAlias	16	0.014820
39	Array	12	0.011115

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 322479

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	84532	26.213180
1	MODIFIES	84532	26.213180
2	UPDATES	56187	17.423460
3	COMMITTED	21990	6.819049
4	CREATES	19759	6.127221
5	HAS_PARENT	12060	3.739778
6	DELETES	11853	3.675588
7	HAS_COMMIT	10995	3.409524
8	HAS_FILE	5641	1.749261
9	RENAMES	3267	1.013089
10	HAS_NEW_NAME	1751	0.542981
11	HAS_TAG	1591	0.493365
12	ON_COMMIT	1591	0.493365
13	HAS_AUTHOR	1249	0.387312
14	DEPENDS_ON	887	0.275057
15	HAS_KEY	668	0.207145
16	HAS_VALUE	668	0.207145
17	CONTAINS	594	0.184198
18	HAS_COMMITTER	370	0.114736
19	OF_TYPE	338	0.104813
20	EXPORTS	309	0.095820
21	REFERENCES	197	0.061089
22	DECLARES	186	0.057678
23	DECLARES_DEV_DEPENDENCY	169	0.052407
24	DECLARES_DEPENDENCY	161	0.049926
25	HAS_MEMBER	103	0.031940
26	HAS_TYPE_ARGUMENT	94	0.029149
27	DECLARES_SCRIPT	91	0.028219
28	RETURNS	82	0.025428
29	COPIES	79	0.024498

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

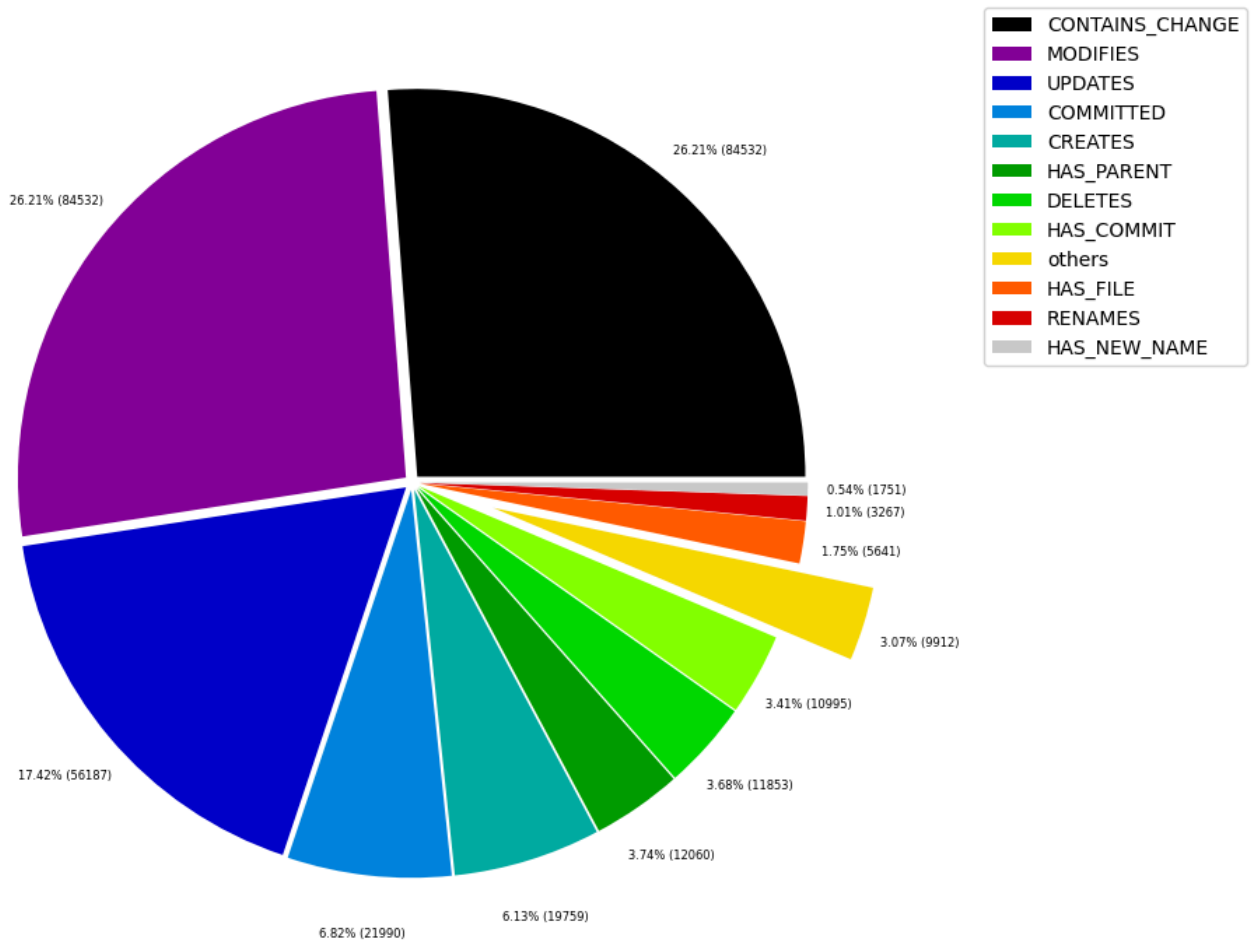


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

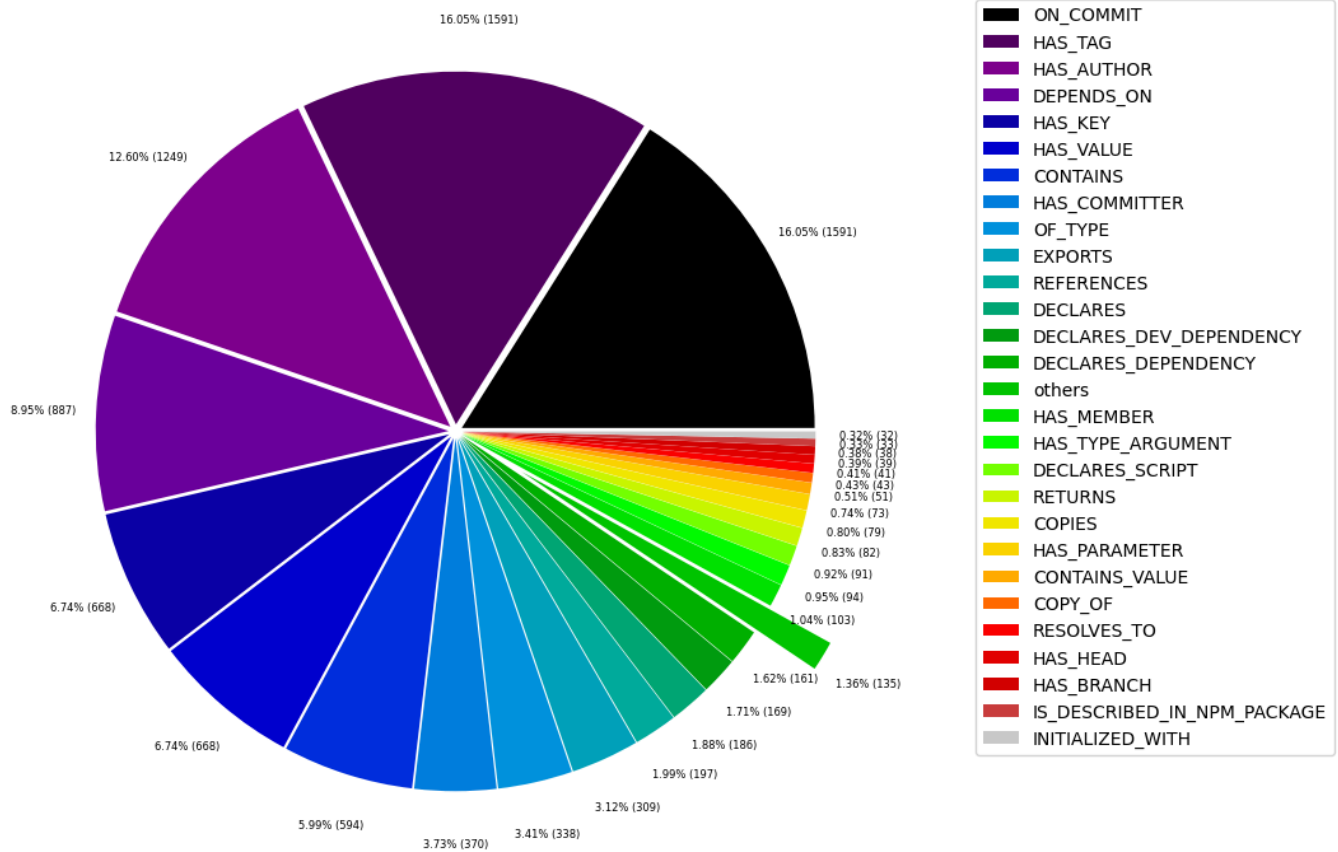
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONSTRAINED_BY	4	0.001240
1	REFERENCED_PROJECTS	5	0.001550
2	MEMBER	6	0.001861
3	HAS_ROOT	6	0.001861
4	HAS_NPM_PACKAGE	6	0.001861
5	HAS_CONFIG	6	0.001861
6	HAS_ARGUMENT	6	0.001861
7	EXTENDS	6	0.001861
8	DECLARES_ENGINE	6	0.001861
9	CONTAINS_PROJECT	6	0.001861
10	CALLS	6	0.001861
11	PARENT	6	0.001861
12	DECLARES_PEER_DEPENDENCY	8	0.002481
13	USES	11	0.003411
14	INCLUDES_CONCEPT	19	0.005892
15	REQUIRES_CONCEPT	28	0.008683
16	INITIALIZED_WITH	32	0.009923
17	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.010233
18	HAS_BRANCH	38	0.011784
19	HAS_HEAD	39	0.012094
20	RESOLVES_TO	41	0.012714
21	COPY_OF	43	0.013334
22	CONTAINS_VALUE	51	0.015815
23	HAS_PARAMETER	73	0.022637
24	COPIES	79	0.024498
25	RETURNS	82	0.025428
26	DECLARES_SCRIPT	91	0.028219
27	HAS_TYPE_ARGUMENT	94	0.029149
28	HAS_MEMBER	103	0.031940
29	DECLARES_DEPENDENCY	161	0.049926

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Change]	MODIFIES	[File, Git]	84532	84532	
1	[Git, Commit]	CONTAINS_CHANGE	[Git, Change]	84532	10995	
2	[Git, Change]	UPDATES	[File, Git]	56187	84532	
3	[Git, Change]	CREATES	[File, Git]	19759	84532	
4	[Git, Commit]	HAS_PARENT	[Git, Commit]	12060	10995	
5	[Git, Change]	DELETES	[File, Git]	11853	84532	
6	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	10995	1	
7	[Author, Git, Person]	COMMITTED	[Git, Commit]	10995	1249	
8	[Committer, Git, Person]	COMMITTED	[Git, Commit]	10995	370	
9	[Repository, File, Git]	HAS_FILE	[File, Git]	5641	1	
10	[Git, Change]	RENAMES	[File, Git]	3267	84532	
11	[File, Git]	HAS_NEW_NAME	[File, Git]	1751	5641	
12	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1591	1	
13	[Git, Tag]	ON_COMMIT	[Git, Commit]	1591	1591	
14	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1249	1	
15	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
16	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
17	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
18	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
19	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
20	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
21	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	
22	[Package, File, Json, NPM]	DECLARES_DEPENDENCY	[NPM, Dependency]	161	29	
23	[Type, TS, Union]	CONTAINS	[Type, TS, Primitive]	147	119	
24	[Type, TS, Declared]	REFERENCES	[TS, ExternalDeclaration]	143	277	
25	[Type, TS, Union]	CONTAINS	[Type, TS, Literal]	119	119	
26	[Json, Key]	HAS_VALUE	[Json, Value, Object]	104	668	
27	[Type, Object, TS]	HAS_MEMBER	[Type, TS, ObjectMember]	102	39	
28	[Package, File, Json, NPM]	DECLARES_SCRIPT	[NPM, Script]	91	29	
29	[Git, Change]	COPIES	[File, Git]	79	84532	

Graph Density

total_number_of_nodes (vertices): 107964

total_number_of_relationships (edges): 322479

-> total directed graph density: 2.7666073600227552e-05

-> total directed graph density in percent: 0.002766607360022755